EAP

Enterprise Analytics Platform

Who am I?

- Georg Rösch
- Senior Data Engineer
- Global IT & Platforms → Platform Engineering → Data Platform
- Former Data&Insights → Team Data Engineering

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EAP Definition

sonnen’s Enterprise Analytics Platform (EAP) is a cloud-based (AWS) unified data and analytics ecosystem covering the full lifecycle—from ingestion and storage to analysis and data products. It provides a shared, governed data foundation that enables self-service analytics, trusted insights, and data-driven decision-making across all teams.

Use Cases

- Smart Monitoring / Trouble Bubble
- Consumption & PV Production Forecasts
- Market Intelligence
- Project Portfolio Management
- Contract Portfolio Managment (Energy Ops)
- Hardware Delivery Report
- Marketing / Sales Reports
- Business Performance
- Case & Failure Analysis
- …

Data Sources

Internal

- SAP
- Salesforce
- AthenaDB
- SpreeDB
- sonnenDB
- Anaplan
- Jira
- Product Engine
- …

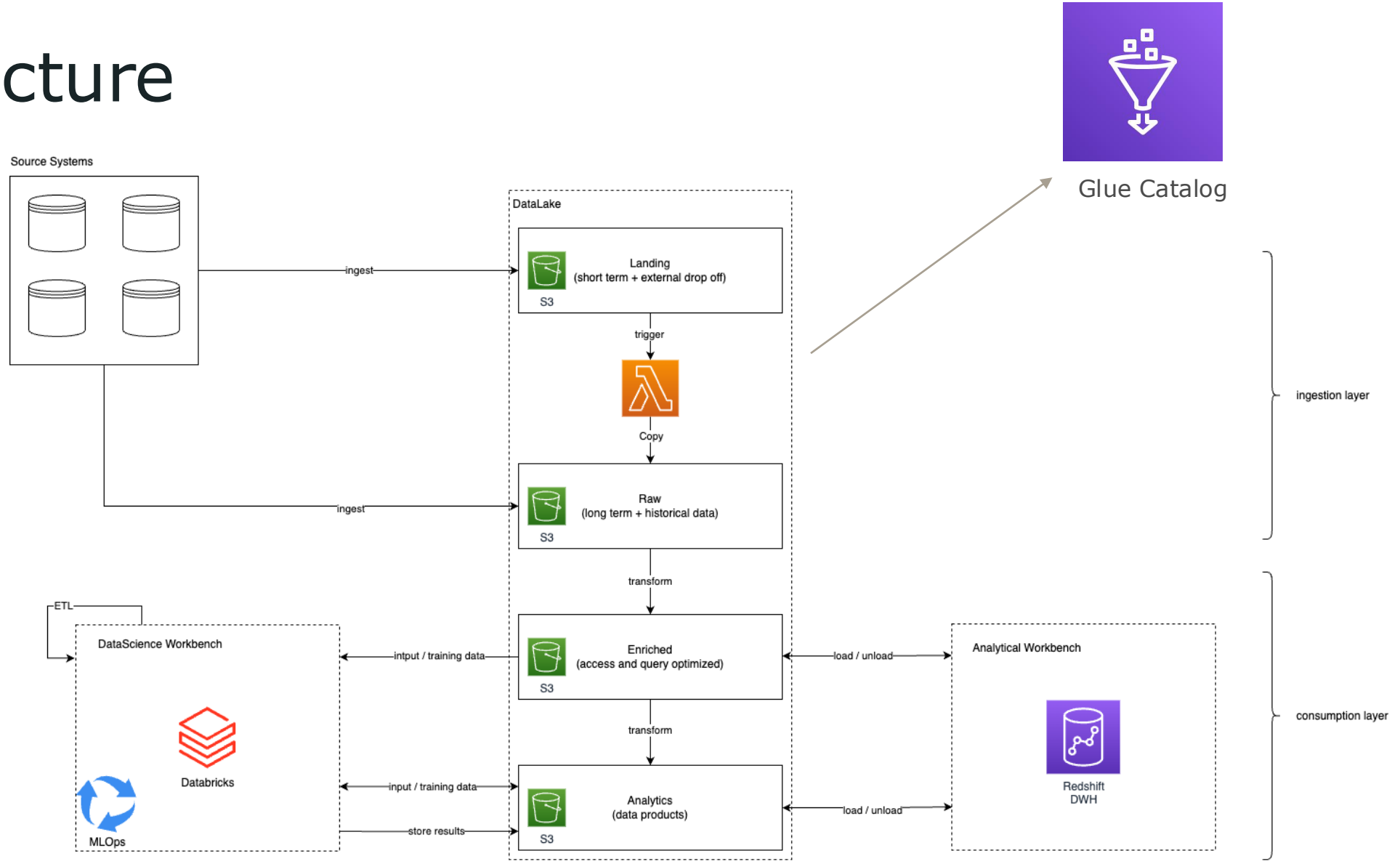
External

- Markstammdatenregister
- Powercloud
- Zuora
- Weather Data (meteomatics)
- Five9
- Google & Apple App Ratings
- …

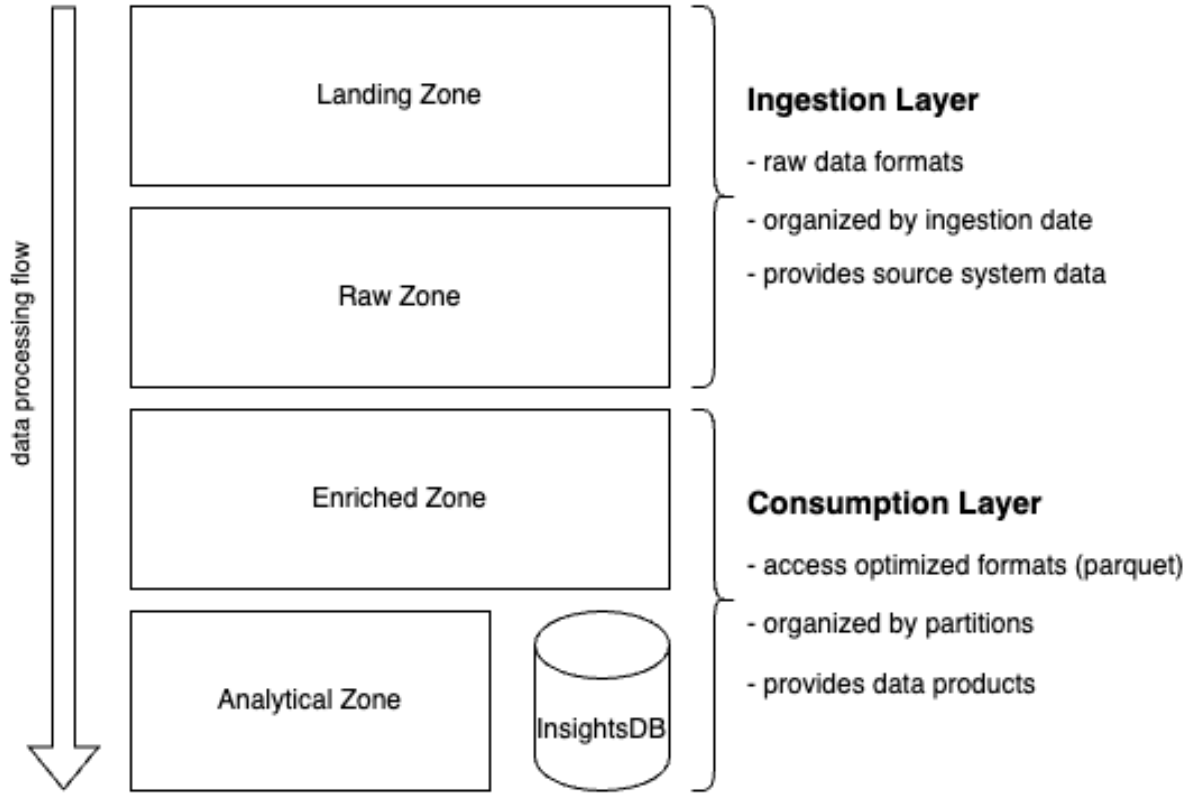
Historization – Slowly Changing Dimensions

Id	Timestamp	My Column	Current
1	2026-03-25T09:05:00+00	value3	True
1	2026-03-24T07:22:00+00	value2	False
1	2026-03-23T14:35:00+00	value1	False

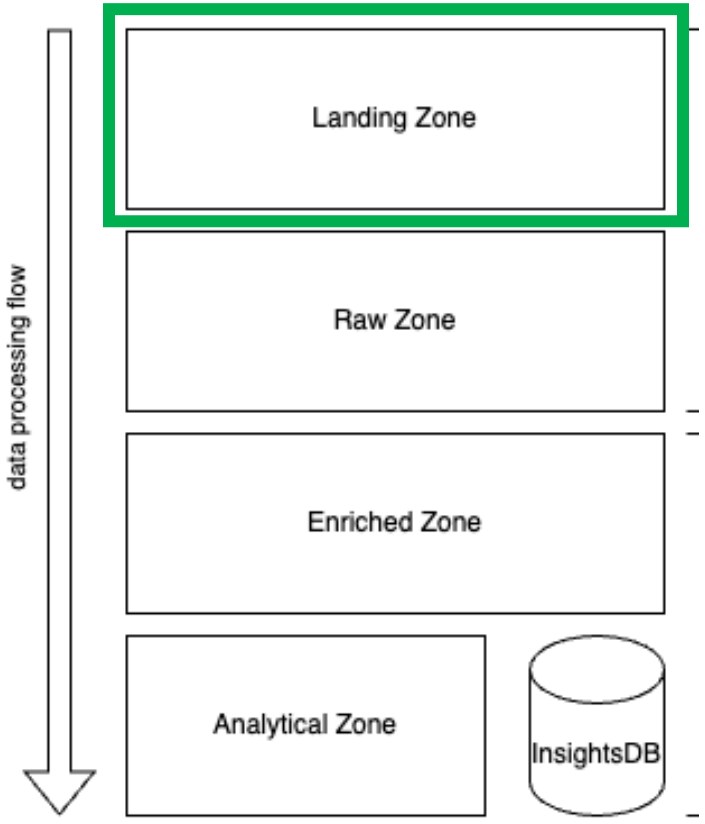
Architecture



Ingestion / Data Lake

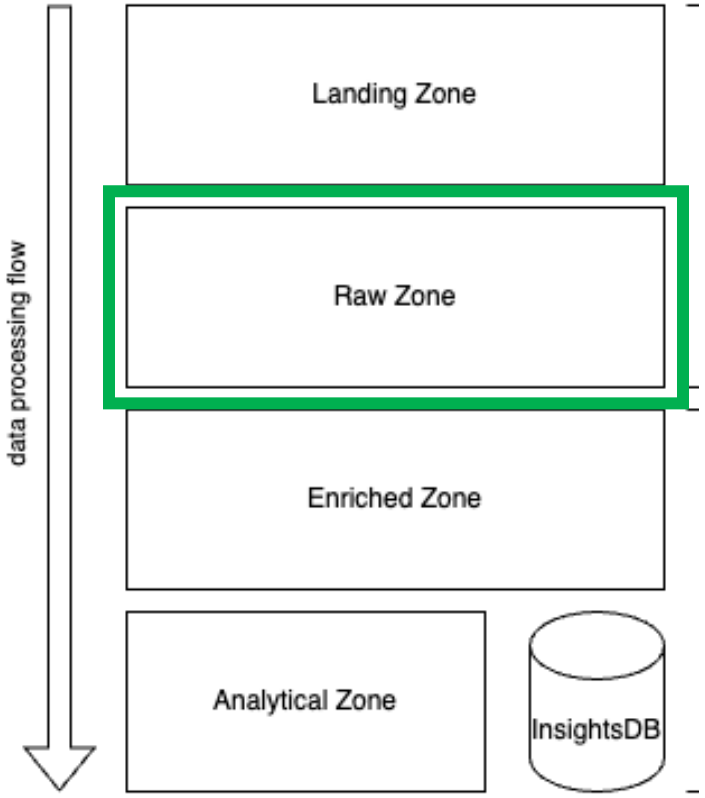


Landing Zone



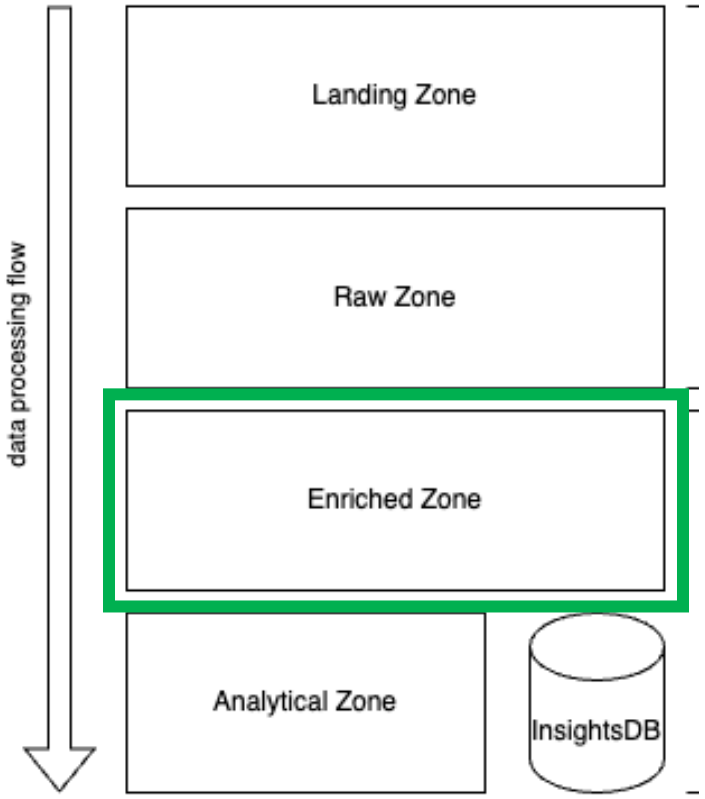
Purpose	•ingestion layer •short term storage •external drop-off zone •data masking / GDPR compliance
Retention time	days
Lifecycle rules	deletion after 1 day
Bucket	bi-*env*-s3-landing
Structure	/drop_off/source/table/filename.filetype

Raw Zone



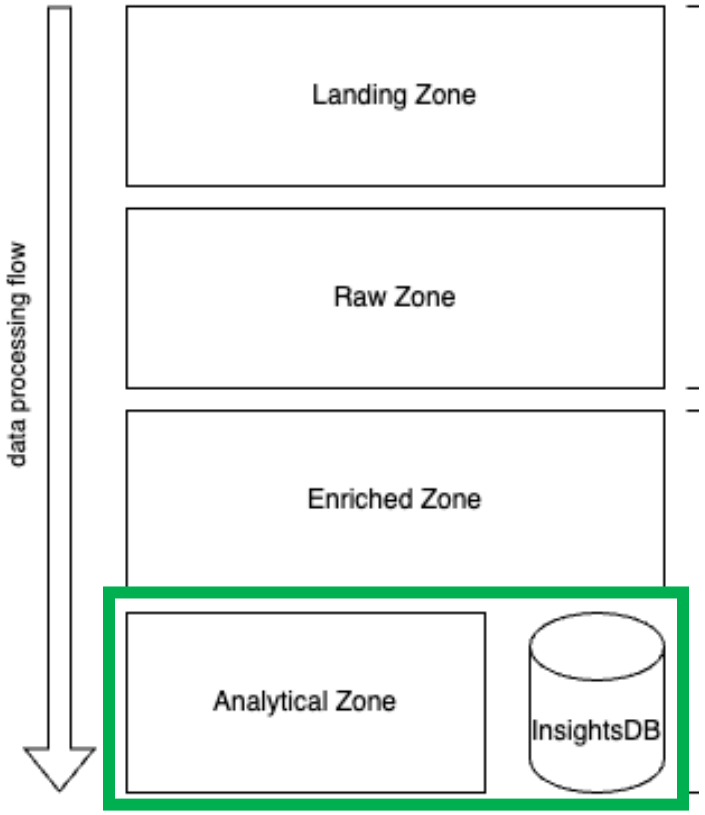
Purpose	•ingestion layer •long term storage •historized raw data from operational systems
Retention time	forever
Lifecycle rules	•versioning enabled •intelligent tiering enabled •after 1 year → Glacier Flexible Retrieval •after 2 years → Glacier Deep Archive
Bucket	bi-*env*-s3-raw
Structure	/source/table/yyyy/mm/dd/table_timestamp.filetype

Enriched Zone



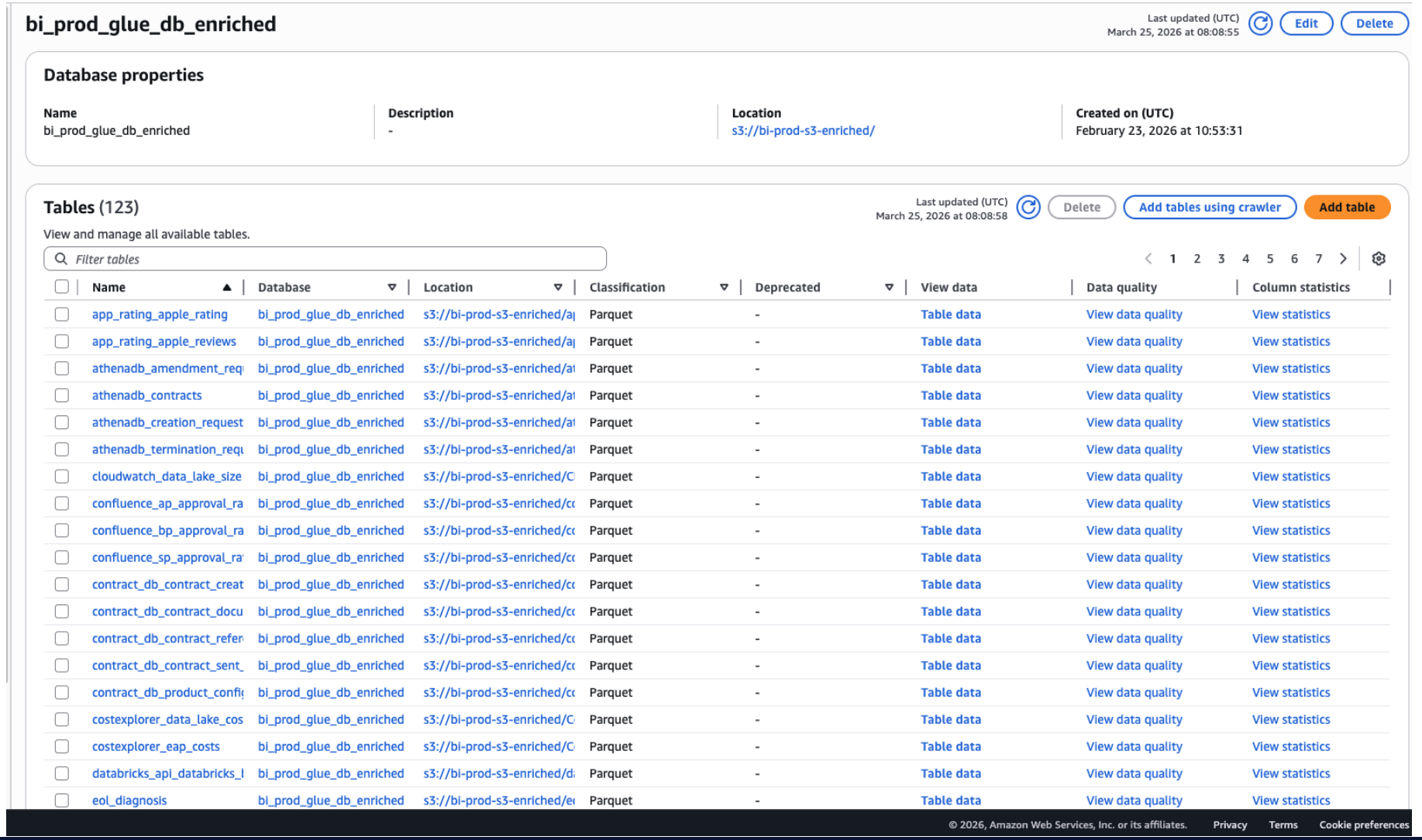
Purpose	•consumption layer •long term storage •optimized for access and query performance •cleansed and quality-checked data •**Parquet and delta tables**
Retention time	forever
Lifecycle rules	•versioning enabled •intelligent tiering enabled •after 1 year → Glacier Flexible Retrieval •after 2 years → Glacier Deep Archive
Bucket	bi-*env*-s3-enriched
Structure	/source/table/partitions/table_yyyymmdd.parquet

Analytical Zone



Purpose	•consumption layer •long term storage •provides data products / aggregated data •**Parquet and delta tables**
Retention time	forever
Lifecycle rules	•versioning enabled •intelligent tiering enabled •after 1 year → Glacier Flexible Retrieval •after 2 years → Glacier Deep Archive
Bucket	bi-*env*-s3-analytics
Structure	/domain/data_product/partitions/table_yyyymmdd.parquet

Metadata Catalog



bi_prod_glue_db_enriched

Last updated (UTC)
March 25, 2026 at 08:08:55

Edit | Delete

Database properties

Name	Description	Location	Created on (UTC)
bi_prod_glue_db_enriched	-	s3://bi-prod-s3-enriched/	February 23, 2026 at 10:53:31

Tables (123)

Last updated (UTC)
March 25, 2026 at 08:08:58

Delete | Add tables using crawler | Add table

View and manage all available tables.

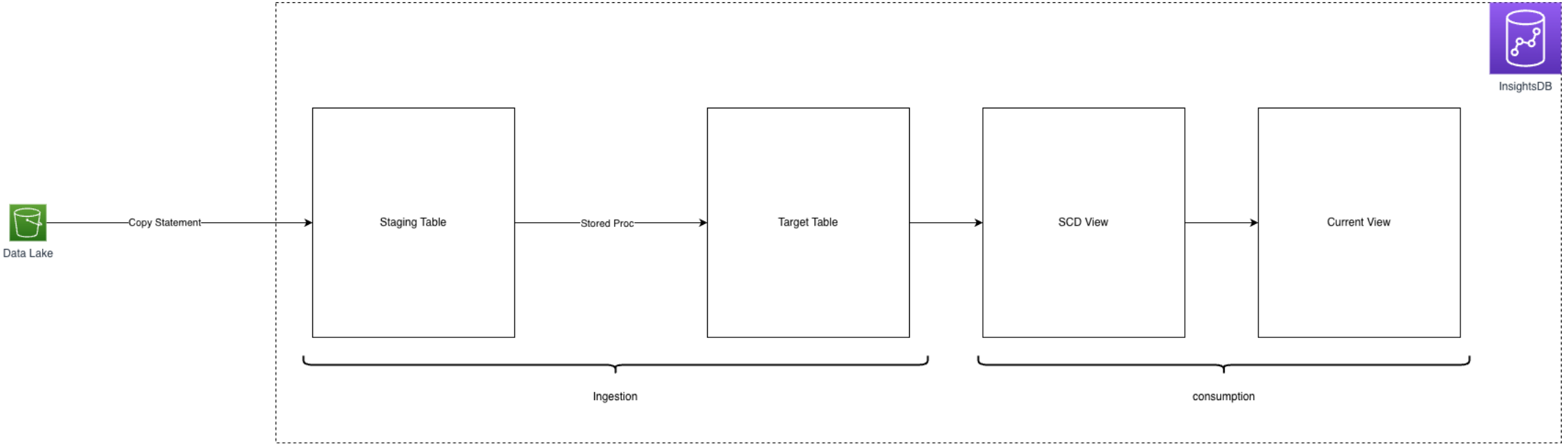
Filter tables

1 2 3 4 5 6 7

Name	Database	Location	Classification	Deprecated	View data	Data quality	Column statistics
app_rating_apple_rating	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
app_rating_apple_reviews	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
athenadb_amendment_req	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
athenadb_contracts	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
athenadb_creation_request	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
athenadb_termination_requ	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/a	Parquet	-	Table data	View data quality	View statistics
cloudwatch_data_lake_size	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/C	Parquet	-	Table data	View data quality	View statistics
confluence_ap_approval_ra	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
confluence_bp_approval_ra	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
confluence_sp_approval_ra	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
contract_db_contract_creat	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
contract_db_contract_docu	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
contract_db_contract_refer	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
contract_db_contract_sent_	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
contract_db_product_confi	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/c	Parquet	-	Table data	View data quality	View statistics
costexplorer_data_lake_cos	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/C	Parquet	-	Table data	View data quality	View statistics
costexplorer_eap_costs	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/C	Parquet	-	Table data	View data quality	View statistics
databricks_api_databricks_l	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/d	Parquet	-	Table data	View data quality	View statistics
eol_diagnosis	bi_prod_glue_db_enriched	s3://bi-prod-s3-enriched/e	Parquet	-	Table data	View data quality	View statistics

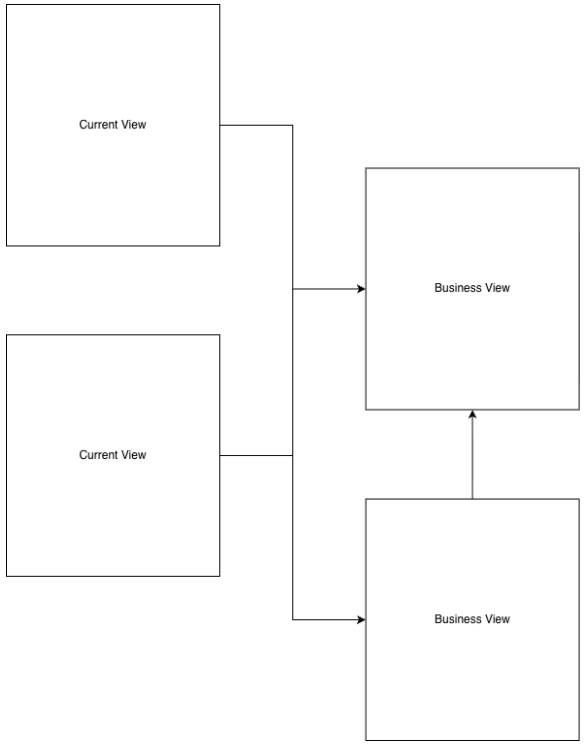
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InsightsDB

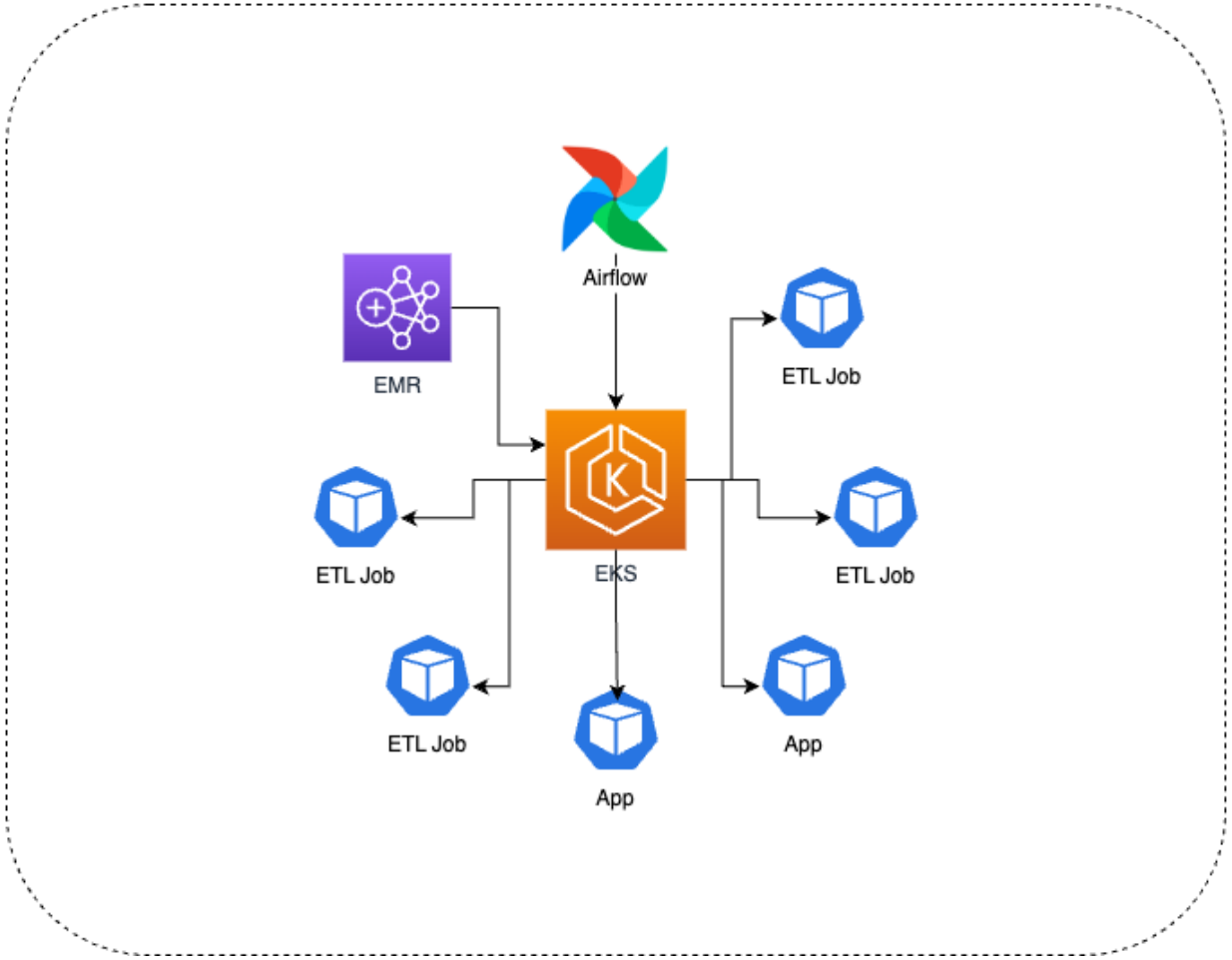

InsightsDB
Data Lake
Copy Statement
Staging Table
Stored Proc
Target Table
SCD View
Current View
Ingestion
consumption

Database Management

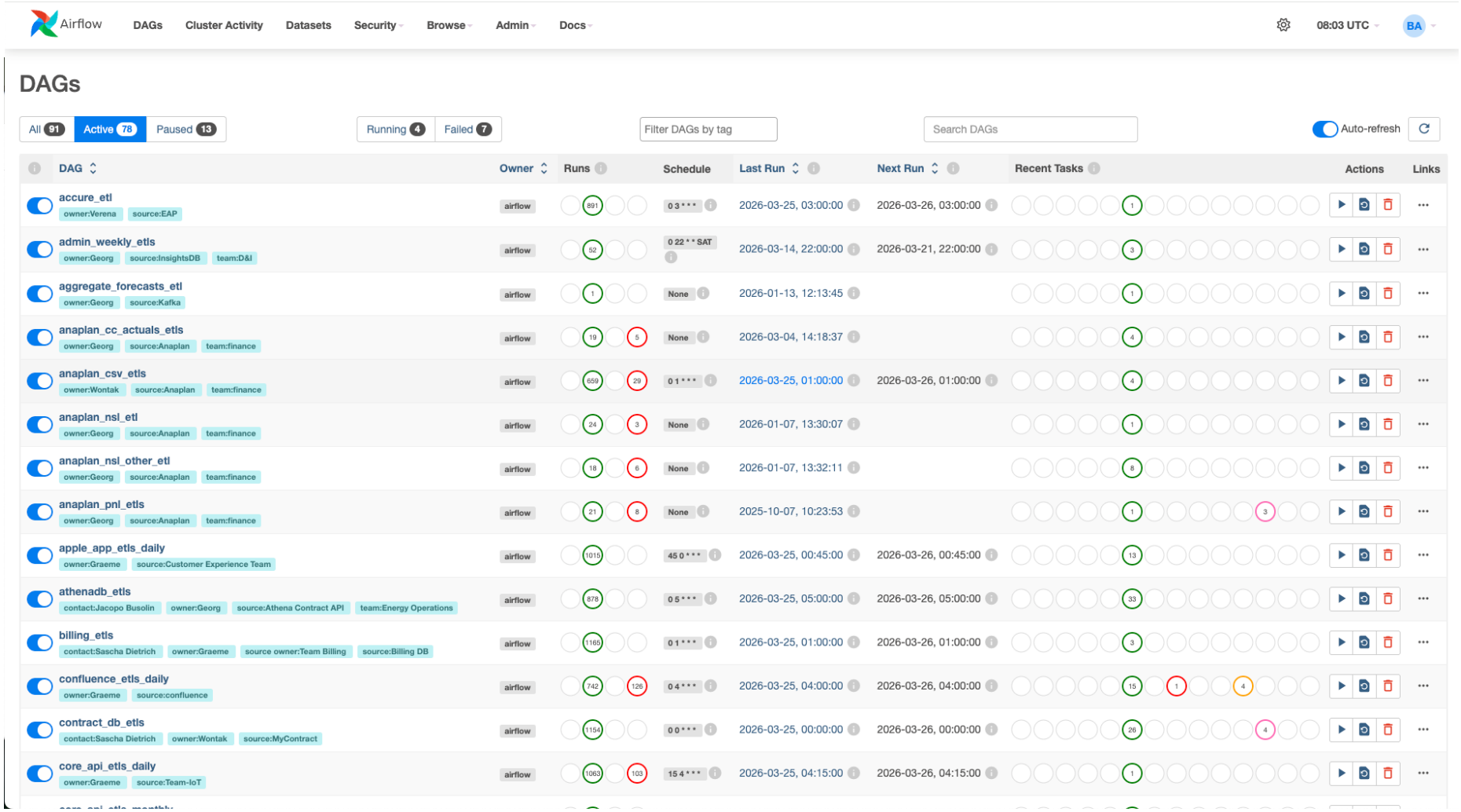
- Changes are tracked in version control
- Enables stakeholders to create PRs
- → code reviews



Orchestration



Airflow

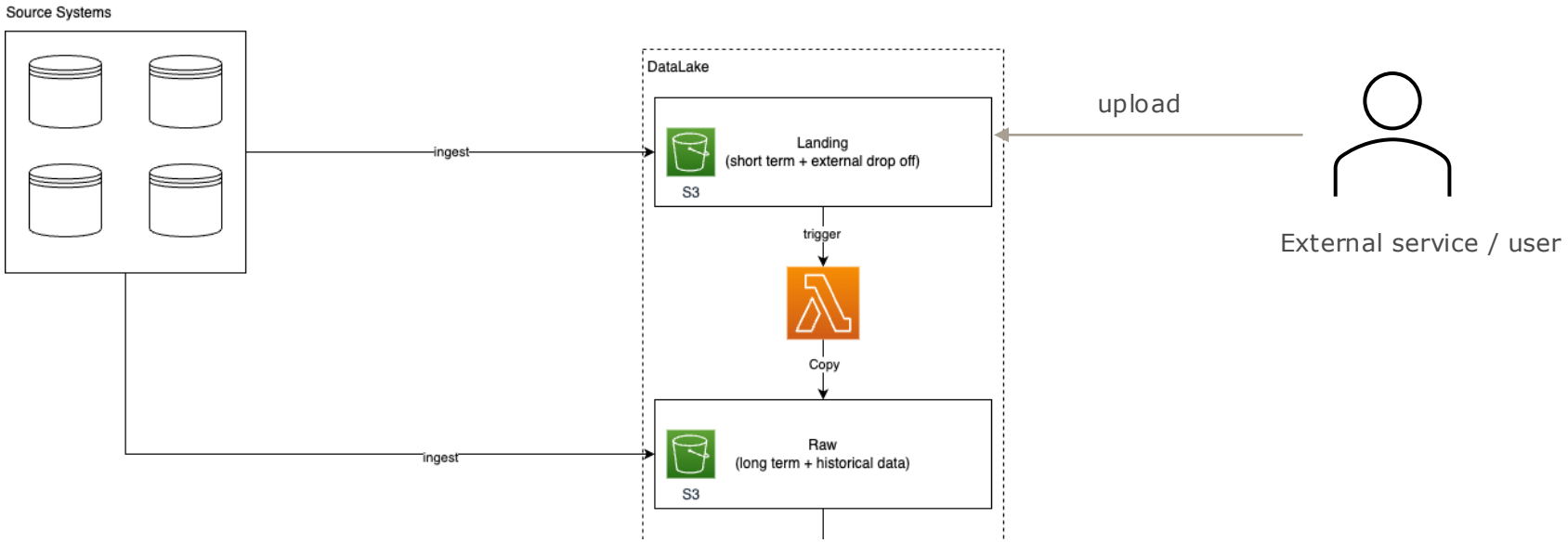

Airflow
DAGs
Cluster Activity
Datasets
Security
Browse
Admin
Docs
08:03 UTC
BA
DAGs
All 91
Active 78
Paused 13
Running 4
Failed 7
Filter DAGs by tag
Search DAGs
Auto-refresh
DAG
Owner
Runs
Schedule
Last Run
Next Run
Recent Tasks
Actions
Links
accure_etl
owner:Verena
source:EAP
airflow
0 3 * * *
2026-03-25, 03:00:00
2026-03-26, 03:00:00
admin_weekly_etls
owner:Georg
source:InsightsDB
team:D&I
airflow
0 22 * * SAT
2026-03-14, 22:00:00
2026-03-21, 22:00:00
aggregate_forecasts_etl
owner:Georg
source:Kafka
airflow
None
2026-01-13, 12:13:45
anaplan_cc_actuals_etls
owner:Georg
source:Anaplan
team:finance
airflow
None
2026-03-04, 14:18:37
anaplan_csv_etls
owner:Wontak
source:Anaplan
team:finance
airflow
0 1 * * *
2026-03-25, 01:00:00
2026-03-26, 01:00:00
anaplan_nsl_etl
owner:Georg
source:Anaplan
team:finance
airflow
None
2026-01-07, 13:30:07
anaplan_nsl_other_etl
owner:Georg
source:Anaplan
team:finance
airflow
None
2026-01-07, 13:32:11
anaplan_pnl_etls
owner:Georg
source:Anaplan
team:finance
airflow
None
2025-10-07, 10:23:53
apple_app_etls_daily
owner:Graeme
source:Customer Experience Team
airflow
45 0 * * *
2026-03-25, 00:45:00
2026-03-26, 00:45:00
athenadb_etls
contact:Jacopo Busolin
owner:Georg
source:Athena Contract API
team:Energy Operations
airflow
0 5 * * *
2026-03-25, 05:00:00
2026-03-26, 05:00:00
billing_etls
contact:Sascha Dietrich
owner:Graeme
source owner:Team Billing
source:Billing DB
airflow
0 1 * * *
2026-03-25, 01:00:00
2026-03-26, 01:00:00
confluence_etls_daily
owner:Graeme
source:confluence
airflow
0 4 * * *
2026-03-25, 04:00:00
2026-03-26, 04:00:00
contract_db_etls
contact:Sascha Dietrich
owner:Wontak
source:MyContract
airflow
0 0 * * *
2026-03-25, 00:00:00
2026-03-26, 00:00:00
core_api_etls_daily
owner:Graeme
source:Team-IoT
airflow
15 4 * * *
2026-03-25, 04:15:00
2026-03-26, 04:15:00

Orchestration

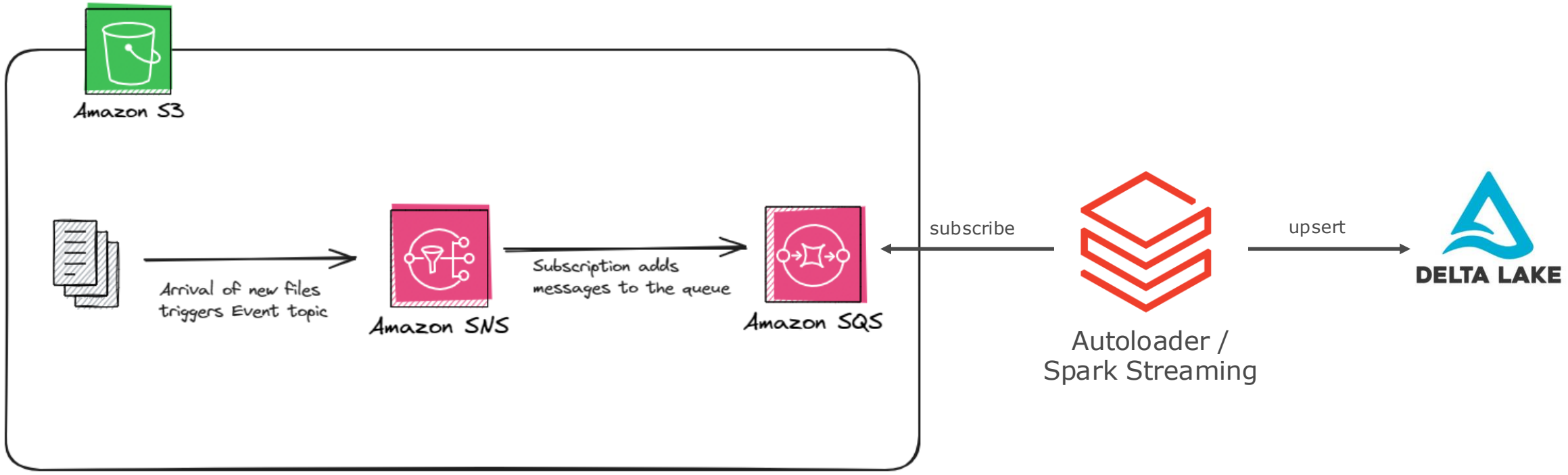


databricks

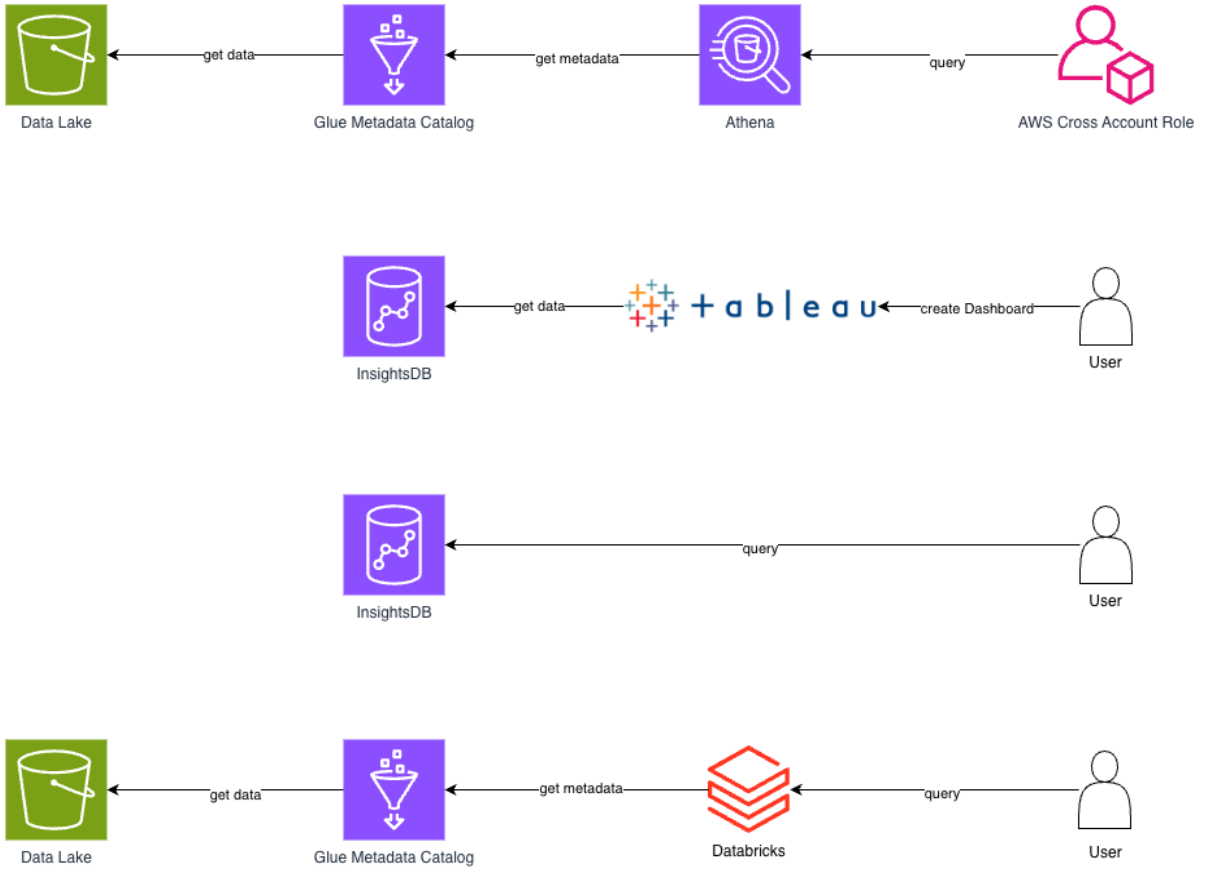
Pull and Event Based Pipelines



Databricks Autoloader



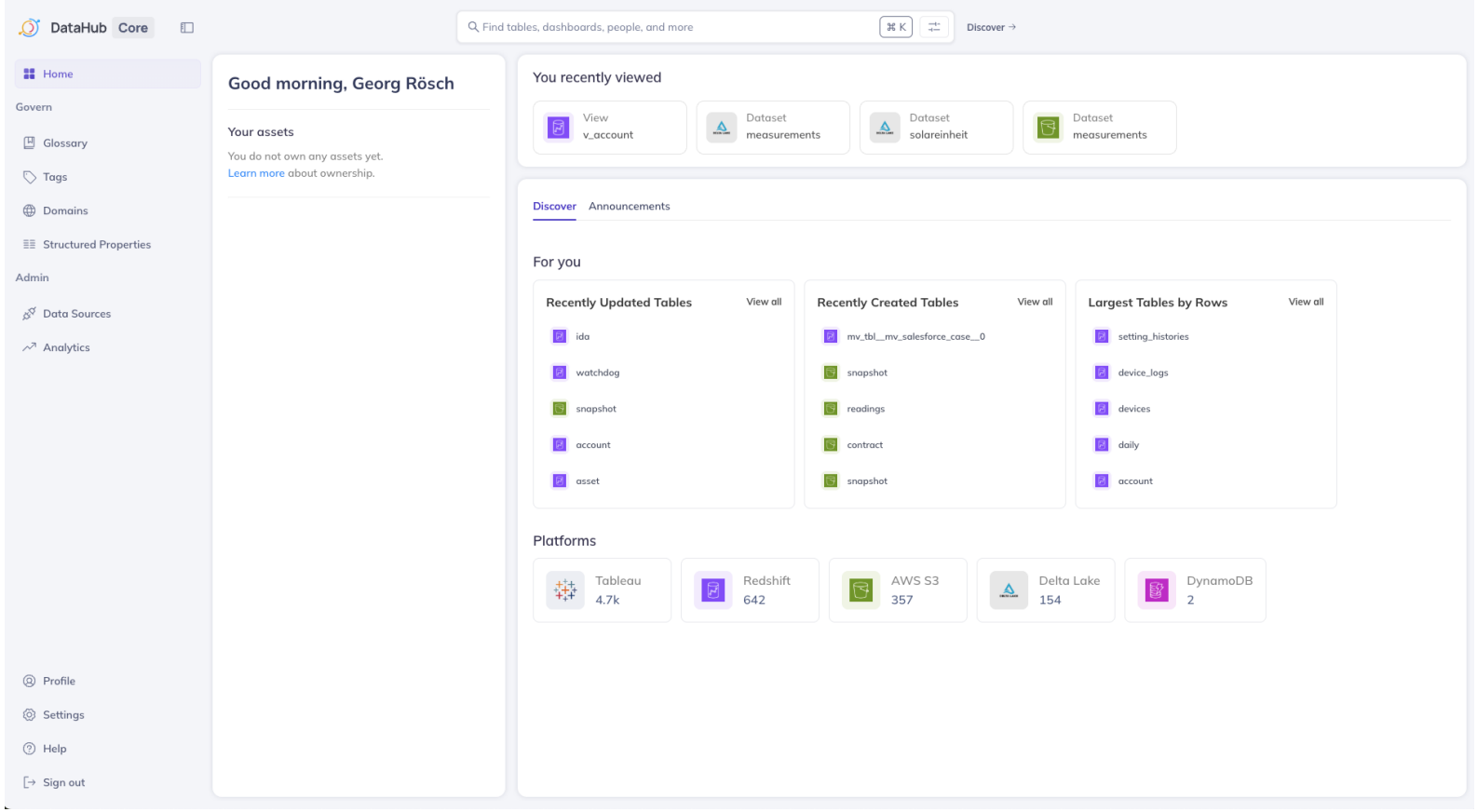
Access


Data Lake
get data
Glue Metadata Catalog
get metadata
Athena
query
AWS Cross Account Role
InsightsDB
get data
tableau
create Dashboard
User
InsightsDB
query
User
Data Lake
get data
Glue Metadata Catalog
get metadata
Databricks
query
User

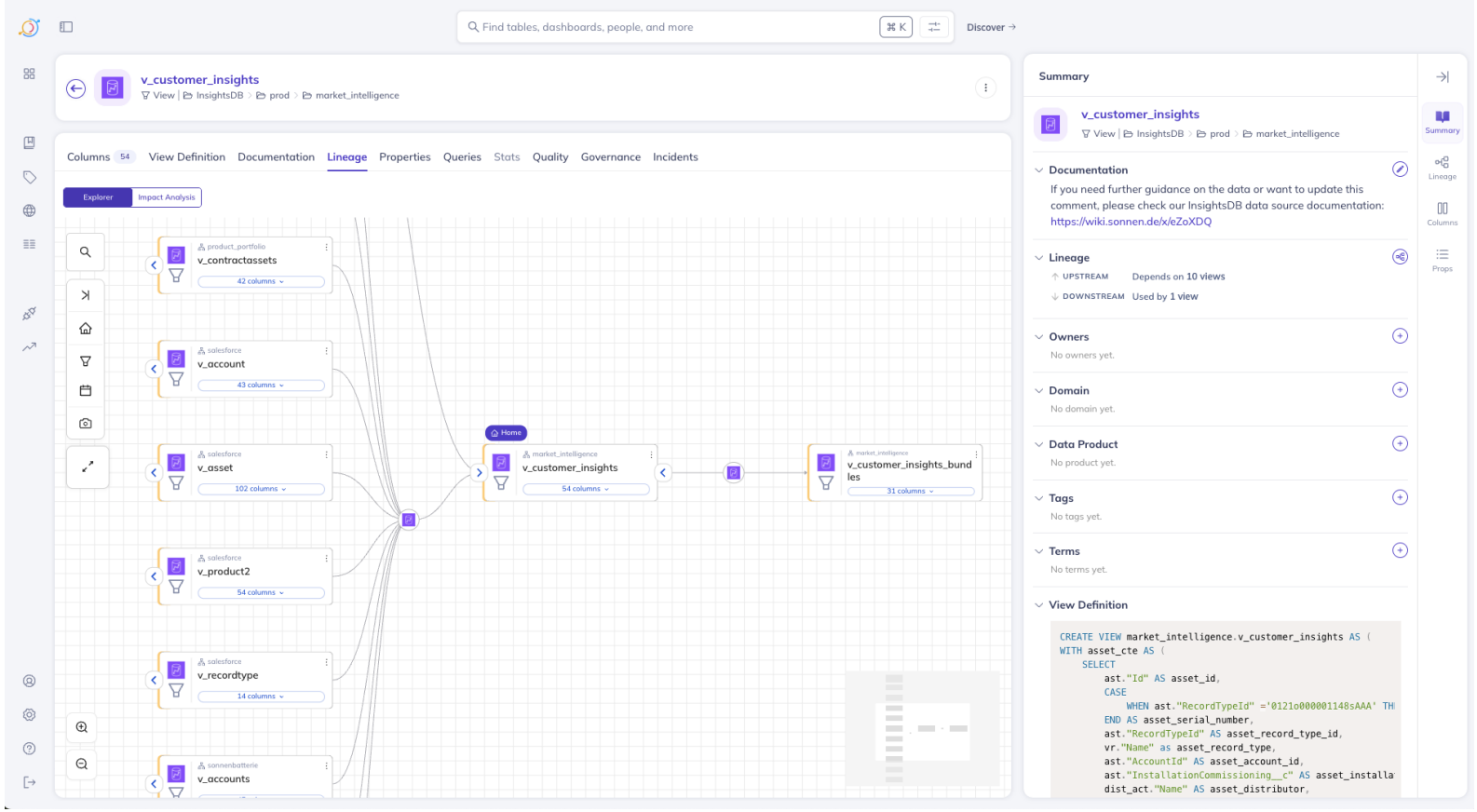
InsightsDB – Redshift Spectrum (coming soon)



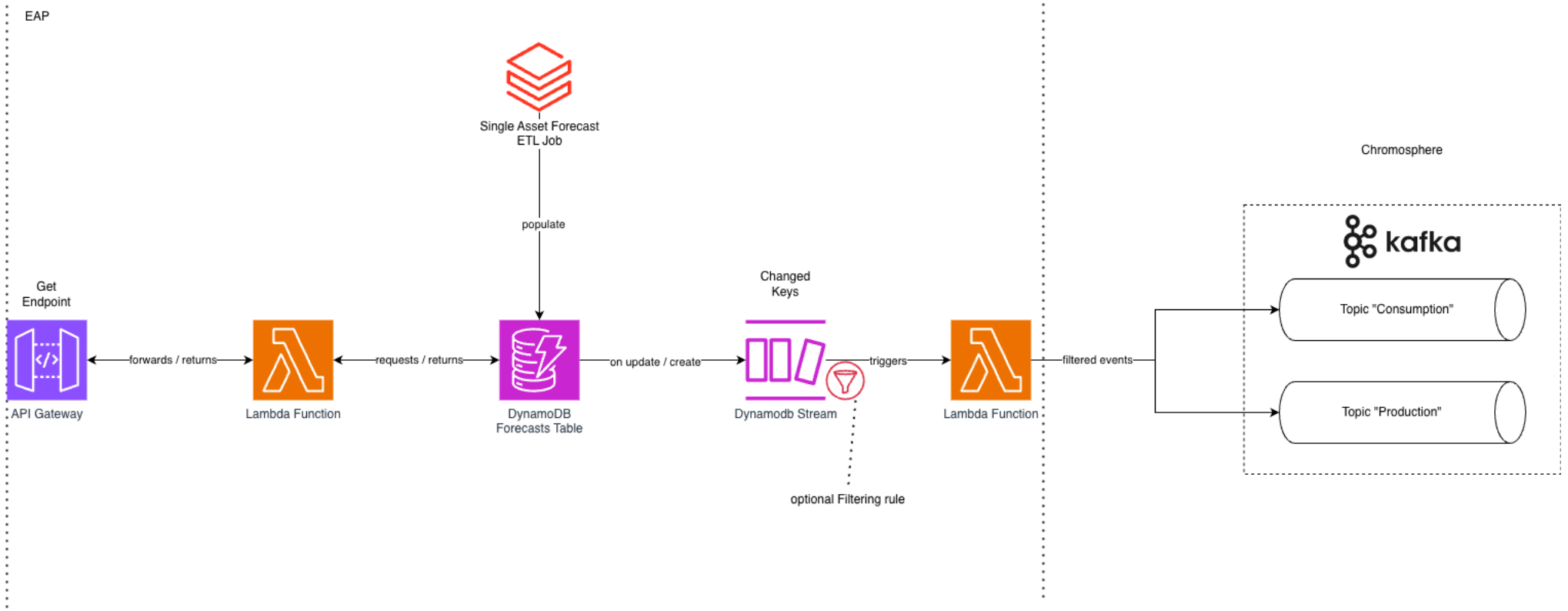
Datahub


DataHub Core
Find tables, dashboards, people, and more
⌘ K
Discover →
Home
Govern
Glossary
Tags
Domains
Structured Properties
Admin
Data Sources
Analytics
Profile
Settings
Help
Sign out
Good morning, Georg Rösch
Your assets
You do not own any assets yet.
Learn more about ownership.
You recently viewed
View v_account
Dataset measurements
Dataset solareinheit
Dataset measurements
Discover
Announcements
For you
Recently Updated Tables
View all
ida
watchdog
snapshot
account
asset
Recently Created Tables
View all
mv_tbl__mv_salesforce_case__0
snapshot
readings
contract
snapshot
Largest Tables by Rows
View all
setting_histories
device_logs
devices
daily
account
Platforms
Tableau 4.7k
Redshift 642
AWS S3 357
Delta Lake 154
DynamoDB 2

Datahub


Find tables, dashboards, people, and more
Discover
v_customer_insights
View | InsightsDB prod market_intelligence
Columns 54
View Definition
Documentation
Lineage
Properties
Queries
Stats
Quality
Governance
Incidents
Explorer
Impact Analysis
v_contractassets
v_account
v_asset
v_product2
v_recordtype
v_accounts
v_customer_insights
v_customer_insights_bundles
Summary
Documentation
If you need further guidance on the data or want to update this comment, please check our InsightsDB data source documentation:
https://wiki.sonnen.de/x/eZoXDQ
Lineage
UPSTREAM Depends on 10 views
DOWNSTREAM Used by 1 view
Owners
No owners yet.
Domain
No domain yet.
Data Product
No product yet.
Tags
No tags yet.
Terms
No terms yet.
View Definition
CREATE VIEW market_intelligence.v_customer_insights AS (
WITH asset_cte AS (
SELECT
ast."Id" AS asset_id,
CASE
WHEN ast."RecordTypeId" ='0121o000001148sAAA' TH
END AS asset_serial_number,
ast."RecordTypeId" AS asset_record_type_id,
vr."Name" as asset_record_type,
ast."AccountId" AS asset_account_id,
ast."InstallationCommissioning__c" AS asset_installa
dist_act."Name" AS asset_distributor,

Forecast Service



Tech Stack

- Python (Pandas, PyArrow, boto3)
- Apache Spark
- Terraform
- Airflow
- EKS
- Redshift
- Lambda
- S3
- Kafka

Thank you!